Three Numbers per Rater: Disagreement in Preference Data Is How People Use the Scale

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Abstract

Pluralistic alignment trains reward models that represent disagreement among people — as a distribution over ratings, as demographic groups, or as a learned user latent. All three assume that what makes people disagree is something a model should learn from the text or from who the people are. We measure, on three public multi-rater datasets (MultiPref pairwise preferences, DICES conversational-safety ratings, PRISM response ratings), where the reproducible structure of disagreement actually sits. Split-half analysis shows that the shape of the rating distribution beyond its location is poorly reproducible across rater panels (split-half reliability 0.25 on MultiPref), that about a tenth of the reproducible part is decodable from a frozen LLM representation of the text, and that rater-side structure dominates item-side dispersion by a factor of 25 in held-out likelihood. Acting on this, we model each rater by a few text-independent offsets on the reward model's output — how they use the scale — learned from a handful of their ratings. For raters never seen in training, offsets fitted on 20 ratings improve held-out log-likelihood over a distributional reward model by 0.08–0.10 nats per rating on all three datasets, while demographic or crowd/expert groups improve it by at most 0.003, a tuned 8-dimensional text-interacting user latent does no better, and reimplementations of two published personalised reward models (VPL, PAL) gain less; on binary preference labels none of the methods personalises at all. On pairwise preferences, a symmetric three-parameter version (how often a rater says "clearly", "slightly", "tie") is the best model at every number of shots and also improves predicting which response the rater prefers; adding an A-vs-B position term from few ratings hurts. What the offsets encode is task-specific: rating intensity on two-sided preference scales, leniency on single-response ratings. On top of a LoRA-fine-tuned reward model the gain is unchanged (+0.100) and adds to the fine-tuning gain, and it exceeds what the same LLM learns from the rater's ratings placed in its prompt (+0.042).

1. Introduction

Reward models and LLM judges are trained on human ratings that disagree. A recent line of work treats this disagreement as signal to be modelled rather than noise to be averaged: distributional preference learning estimates a distribution over each alternative's utility rather than a point score (Siththaranjan et al., 2024); jury learning predicts individual annotators from content, annotator and demographic-group embeddings (Gordon et al., 2022), and large datasets such as PRISM are analysed by sociodemographic group (Kirk et al., 2024); personalised reward models learn a per-user latent, inferred for a new user from a few examples (Poddar et al., 2024; Chen et al., 2025). The pluralistic-alignment agenda (Sorensen et al., 2024) makes the representation of disagreement a first-class goal.

These approaches agree on one premise: the disagreement worth modelling is a property of the *item* (some responses are contested) or of *who the raters are* (groups and individuals hold different values about content). There is a third, older possibility that measurement theory takes for granted and reward modelling largely does not: raters differ in **how they use the response scale** — some say "clearly better" where others say "slightly better", some rarely call a tie, some are harsher across the board (Linacre, 1989; Baumgartner & Steenkamp, 2001). Such response styles are text-independent, cheap to estimate, and invisible to a model that only conditions on content or demographics.

We ask two questions. **(Q1, measurement)** In public multi-rater preference and rating data, how much of the disagreement is reproducible across raters, how much of the reproducible part is in the text, and how much is rater-side? **(Q2, modelling)** If the answer is "mostly rater-side", what is the cheapest rater model that captures it for a *new* rater from

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few ratings — and how does it compare with distributional, group-conditioned and low-rank personalised reward models under the same protocol?

Contributions. (1) A learner-free measurement of disagreement in preference data — split-half reliability of the distribution's shape, cross-fitted text decodability, and a rater–item decomposition of held-out likelihood — showing that rater-side structure dominates (§3). (2) A factorised reward model: a text model for the item, and a few text-independent offsets per rater on its output, shrunk toward the population and adapted from k ratings (§4). (3) A new-rater, few-shot evaluation on three datasets against distributional, group-conditioned and low-rank-latent reward models, with all hypotheses, amendments and failures pre-registered and reported (§5): offsets beat groups everywhere, match a tuned low-rank latent, and a symmetric three-number style model is best on pairwise preferences. (4) The observation that what "style" means depends on the scale: intensity on two-sided preferences, leniency on single-response ratings (§5.3).

2. Related work

Pluralistic and distributional reward models. Distributional preference learning (Siththaranjan et al., 2024), diverging-preference detection (Zhang et al., 2025), PRISM (Kirk et al., 2024), MultiPref (Miranda et al., 2025) and DICES (Aroyo et al., 2023) motivate modelling disagreement; jury learning (Gordon et al., 2022) and multi-annotator models (Mostafazadeh Davani et al., 2022) predict individual annotators. **Personalised reward models** infer a user latent from a few examples (Poddar et al., 2024; Chen et al., 2025) or learn per-user embeddings jointly with the reward model (Li et al., 2024). **Annotator demographics** as predictors of labels have been found weak in several studies (e.g. Orlikowski et al., 2023). **Measurement theory**: rater severity and scale use are standard facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011) and response styles are a classical survey-methods topic (Baumgartner & Steenkamp, 2001). We connect the last line to the first: for new-rater prediction, the measurement-theory rater facet is what the reward model is missing.

3. Where does disagreement in preference data sit? (Q1)

Data. MultiPref (Miranda et al., 2025; 10,461 comparisons; 4 ratings each by 2 crowd and 2 expert raters; 227 persistent raters; 5 ordered levels A-clearly … B-clearly). PRISM (Kirk et al., 2024; 68,371 responses each rated 1–100 by one of 1,396 users; binned to 5 fixed levels). DICES-350 (Aroyo et al., 2023; 350 conversations fully crossed with 104 quality-checked raters; safe / unsure / unsafe).

Reproducibility. With the split-half estimator of a companion study (anonymised), the reliability of the rating distribution's shape beyond its exact mode and mean is ρ = 0.25 [0.22, 0.29] on MultiPref overall preference with three raters (helpful 0.29, harmless 0.21): the raw few-rater distribution forecasts a new rater worse than its own location ($\rho < \frac{1}{2}$). **Text decodability.** A cross-fitted ridge decoder on frozen Qwen2.5-7B-Instruct features, scored on an independent rater panel, recovers G/X = 10.5 % [5.7 %, 11.9 %] of the reproducible shape (within-cell permutation control ≤ 0; positive control 0.35) — more than on ordinal quality ratings ($\leq$ a few percent) and less than on NLI (17–26 %). **Rater vs item.** A long-format ordered-probit decomposition with real rater identities and penalties tuned by inner cross-validation: rater-specific thresholds improve held-out likelihood by +0.036 nats per rating over item location alone, item-specific dispersion by a further +0.0014 (ratio 25); of the rater part, rater scale contributes +0.031 and rater shift +0.005.

The reproducible part of preference disagreement is therefore mostly rater-side, and mostly about scale use.

4. A factorised reward model (Q2)

Let $\ell(x) \in \hat{R}K$ be the class logits of a text-only distributional reward model for item x (K ordered rating levels). For rater r we add a text-independent offset vector: $p(y \mid x, r) = \text{softmax}(\ell(x) + b_r)$. Variants: **DF** ($b_r \in \hat{R}K$, free), **DFs** (two-sided preference scales only: b_r tied symmetrically, (e, s, t, s, e) — extreme, slight, tie — 3 parameters, no A-vs-B position term). Rater offsets have a Gaussian prior centred on the training-rater mean; a new rater's offsets are the MAP estimate from k of their ratings with the text model frozen. Baselines under the same protocol: **D** (text-only distributional RM), **DG** (group offsets: crowd/expert, race, region), **DL** (low-rank text-interacting latent: $\ell(x) + (Wx + w_0)\cdot z_r$, d = 8, the personalised-RM family). An interpretable probit family (rater shift / log-scale on an ordered-probit text head) is reported as a secondary analysis.

5. Experiments

Protocol. 5 folds; in each, 20 % of raters (stratified by group) and 20 % of prompt groups are held out; models are trained on training raters $\times$ training items with penalties chosen on a validation split; each test rater is adapted on $k \in \{0, 5, 10, 20, 50\}$ of their ratings of *training* items (5 random draws) and evaluated on their ratings of *held-out* items. Primary metric: held-out log-likelihood per rating; also RPS and preference-direction accuracy; rater-cluster bootstrap (B = 2,000). Text model in §5.1–5.3: frozen Qwen2.5-7B features with a ridge head; §5.4: a LoRA-fine-tuned reward model.

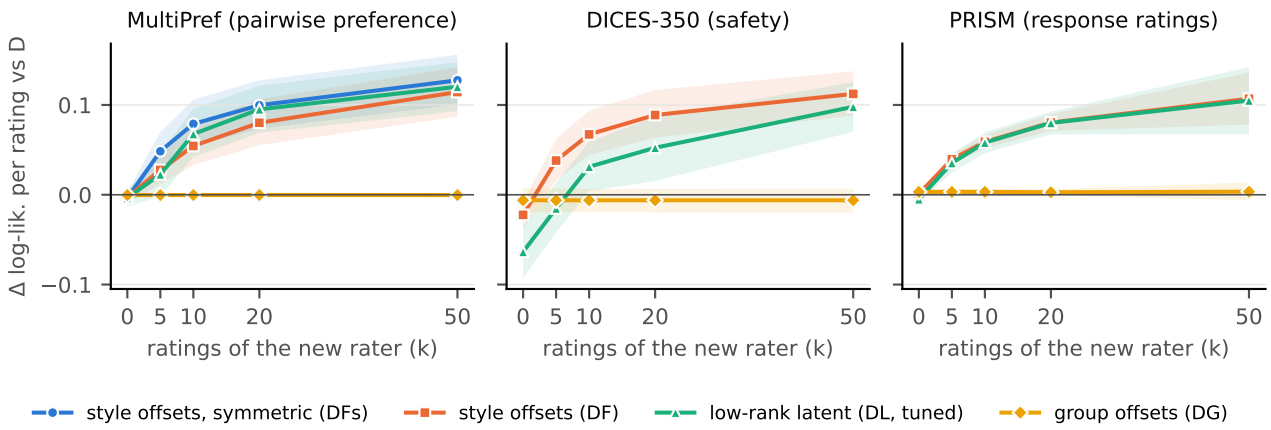


Figure 1. New raters: gain in held-out log-likelihood per rating over the text-only distributional reward model (D), as a function of the number k of the new rater's ratings used for adaptation; 95% rater-cluster bootstrap bands, 5 folds. Text-independent rater offsets (DF; symmetric three-parameter DFs on two-sided preferences) gain 0.08–0.10 nats per rating at k = 20 on all three datasets; group offsets (crowd/expert, race, region) gain nothing; a low-rank text-interacting rater latent with a few-shot-tuned prior is at best equal.

5.1. Main result (MultiPref)

Figure 1 shows the gain over the text-only reward model as a function of k on all three datasets. | k | DF − D | DFs − D | DG − D | DL − D (tuned prior) | DFs − DL (tuned) | |—|—|—|—|—|—| | 5 | +0.028 [+0.012, +0.044] | +0.048 [+0.029, +0.070] | ≈ 0 | +0.023 [−0.002, +0.047] | +0.026 [+0.016, +0.037] | | 20 | +0.080 [+0.055, +0.106] | **+0.100 [+0.074, +0.128]** | ≈ 0 | +0.095 [+0.069, +0.122] | +0.0045 [−0.0004, +0.0092] | | 50 | +0.114 [+0.087, +0.142] | +0.128 [+0.101, +0.156] | ≈ 0 | +0.120 | +0.007 [+0.002, +0.013] | Preference-direction accuracy (D: 0.674): DFs +0.005 [+0.001, +0.008] at k = 20; DF −0.014 [−0.020, −0.007]. A three-number style model gains as much as a tuned 8-dimensional text-interacting latent and more at k ≤ 10; fitting a rater's A-vs-B position from few ratings costs direction accuracy.

5.2. Replications (k = 20)

5.3. What the offsets encode

In the probit family, rater *scale* matters more than rater *shift* on MultiPref (+0.036 [+0.020, +0.057] at k = 20), but rater *shift* dominates on DICES (−0.053) and PRISM (−0.031): leniency/strictness on single-response rating scales, intensity on two-sided preference scales.

5.4. On a fine-tuned reward model

With a LoRA-fine-tuned Qwen2.5-7B reward model (D-LoRA, trained per fold on the training raters' rating distributions) as the text model, the three-number style model still adds +0.100 [+0.071, +0.130] nats per rating at k = 20 — exactly its gain on the frozen-feature head. Fine-tuning improves the text-only model by +0.027 [+0.016, +0.039] (direction accuracy +1.4 points) and the style model by the same +0.027: the two gains add. Personalising the frozen-feature model with 20 ratings per rater gains more than fine-tuning the text model (+0.072 [+0.041, +0.106]). DFs keeps the direction accuracy of D-LoRA (+0.002 [−0.002, +0.006]). Against in-context personalisation — the same LoRA recipe trained with ten of the rater's other ratings in the prompt and evaluated with the new rater's k ratings in the prompt (two folds, 86–90 raters) — the three offsets on D-LoRA are better at k = 5 (+0.041 [+0.016, +0.065] nats per rating) and at k = 20 (+0.042 [+0.020, +0.065]), and more accurate on which response the rater prefers (+2.9 points); the in-context model learns from the history (+0.048 over D-LoRA) but loses 3.0 points of direction accuracy, at several thousand prompt tokens per prediction.

5.5. Published personalised reward models

We reimplemented VPL (Poddar et al., 2024) and PAL-B (Chen et al., 2025) from their official code as heads on the same frozen per-response embeddings and evaluated them under the same new-rater protocol, refitting the text-only and style models on those embeddings. On graded labels, at k = 20 the style model is better than VPL by +0.092 [+0.068, +0.119] nats per rating and than PAL-B by +0.158. VPL's latent does carry rater information once its posterior collapse is prevented (KL weight selected as 0 or 0.001; +0.087 over its own unpersonalised version), but only lifts it to the level of the text-only model; PAL-B's few-shot user weights give no gain. Their user representations are unrelated to response style (cross-validated $R^2 \approx 0$ against the style offsets). On

dataset	DF − D	DG − D (groups)	DF − DL (tuned prior)	direction acc. DF − D
DICES-350 (safety)	+0.089 [+0.064, +0.117]	−0.006 (race)	+0.037 [+0.022, +0.053]	+0.048 [+0.025, +0.076]
PRISM (ratings)	+0.080 [+0.071, +0.090]	+0.003 (region)	−0.001 [−0.006, +0.003]	+0.005 [+0.001, +0.009]

binary labels — the setting in which both methods were proposed — no method personalises (all gains ≤ 0.002 nats): the individual differences that are predictable for new raters live in the graded levels and ties that binary preference data discard. VPL's per-response reward head is slightly better at predicting *which* response is preferred (0.677 vs 0.668), an effect present without personalisation. The offsets are also a plug-in: added to VPL (fitted on the same k ratings, VPL frozen) they improve it by +0.088 [+0.066, +0.111] nats per rating at k = 20, tie with the style model on likelihood (−0.004 [−0.014, +0.005]) and keep VPL's direction accuracy (+1.1 points over the style model alone).

6. Pre-registration and what did not hold

All analyses followed written protocols; changes after results are logged as amendments with reasons. Not supported or not robust: (i) the 5-parameter DF beats the low-rank latent (H3) holds with the training-rater prior but not with a few-shot-tuned prior on MultiPref (−0.015) and only as a tie on PRISM; (ii) "scale beats shift" (H4) holds only on MultiPref. Two baseline defects found during the study were fixed in favour of the baselines (a missing constant column in the low-rank latent; a scale non-identifiability in the probit group model) and the affected runs are reported.

7. Limitations

Frozen-feature text heads for the main tables (a fine-tuned head in §5.4); three datasets, one language; PRISM scores binned to 5 levels; rater offsets are text-independent by design — interactions with content exist in principle and are what the low-rank latent can capture; we test linear decoders for the text-side measurement only; adaptation uses the rater's ratings of other items, not of the same item.

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